Cyclic Schedule & Protection Behavior — User Guide

1. What is a Cyclic Schedule?

A cyclic schedule repeatedly turns the device ON and OFF within a defined time window. You configure:

- **Start Time** and **End Time** — the overall schedule window
- **ON Duration** — how long the device stays ON in each cycle
- **OFF Duration** — how long the device stays OFF before the next cycle begins

The device automatically repeats these ON/OFF cycles until the schedule window ends.

Example 1:

```
Schedule: 21:00 to 22:00 | ON: 10 min, OFF: 50 min

  21:00 ─── ON ──► 21:10 ─── OFF ──► 22:00 (window ends)
  (Only 1 cycle fits in this window)
```

Example 2:

```
Schedule: 09:00 to 12:00 | ON: 30 min, OFF: 30 min

  09:00 ── ON ──► 09:30 ── OFF ──► 10:00 ── ON ──► 10:30 ── OFF ──► 11:00 ── ON ──►
11:30 ── OFF ──► 12:00
  Cycle 1                         Cycle 2                         Cycle 3
```

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2. Manual Stop During a Cyclic Schedule

When you manually stop the device (via physical button, app, cloud, or webhook) during a cyclic schedule:

> ***The entire schedule is stopped — not just the current cycle.***

All remaining cycles within that schedule window are cancelled. The device will not turn ON again until the **next schedule occurrence** (e.g., the next day).

Example:

```
Schedule: 09:00–12:00 | ON: 30 min, OFF: 30 min

  09:00 ── ON ──► 09:15  User presses STOP
                          ✗ Cycle 2 at 10:00 — CANCELLED
                          ✗ Cycle 3 at 11:00 — CANCELLED
                          ✓ Next day 09:00 — Starts normally
```

Why? Manual stop is treated as an intentional user decision. The device respects it for the remainder of that schedule window.

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3. How Faults Affect a Cyclic Schedule

When a protection fault occurs during a cyclic ON period, the device turns OFF immediately. What happens next depends on the **type of fault**.

Faults That Block the Entire Schedule

These faults are considered serious — they indicate a load or wiring issue that is unlikely to resolve on its own:

Fault	What It Means
Dry Run	Current is too low (e.g., pump running dry)
Overcurrent	Current exceeds the safe limit

Behavior: The device stays OFF for the **remainder of the schedule window**. No further cycles will run.

With Auto-Retry enabled: If you have configured auto-retry for these faults, the device will attempt to restart after the configured retry delay. This can allow a later cycle to run if the retry delay has passed.

Example — Dry Run, no auto-retry:

```
Schedule: 09:00–12:00 | ON: 30 min, OFF: 30 min

  09:00 ── ON ──► 09:15  Dry Run detected → device OFF
                         ✗ Cycle 2 at 10:00 — BLOCKED
                         ✗ Cycle 3 at 11:00 — BLOCKED
                         ✓ Next day 09:00 — Starts normally
```

Example — Dry Run, auto-retry after 5 minutes:

```
  09:00 ── ON ──► 09:15  Dry Run detected → device OFF
                  09:20  Retry window opens
                         ✓ Cycle 2 at 10:00 — CAN START
```

Faults That Clear Between Cycles

These faults are considered transient — the condition may resolve on its own:

Fault	What It Means
Low Voltage	Supply voltage is below the safe limit
High Voltage	Supply voltage is above the safe limit
Phase Failure	One or more phases are missing
Phase Sequence Reverse	Phase wiring is reversed
Float Alert	Float switch has triggered

Behavior: The device turns OFF, but the fault **automatically clears at the next cycle boundary**. The next cycle can start normally — provided the fault condition has actually resolved (e.g., voltage has returned to normal). If the fault persists, the device will detect it again and turn OFF.

Example — Low Voltage:

```
Schedule: 09:00–12:00 | ON: 30 min, OFF: 30 min
```

```
09:00 ── ON ──► 09:15  Low Voltage detected → device OFF
                 09:30  Cycle boundary — fault clears automatically
                        ✓ Cycle 2 at 10:00 — CAN START (if voltage is normal)
```

Fault Detection Timing

Fault	How It Is Detected
Dry Run	Current stays below limit for 4 seconds (checked after 7-second settle time)
Overcurrent	Current stays above limit for 4 seconds (checked after 7-second settle time)
Low Voltage	Voltage stays below limit for 7 seconds
High Voltage	Voltage stays above limit for 7 seconds

The settle time ensures the device ignores brief transients (e.g., inrush current at startup).

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4. MaxRuntime and BreakTime

These are **global safety settings** that apply to all schedule types, including cyclic.

MaxRuntime

The maximum duration the device is allowed to stay ON continuously. If the device has been ON for this long, it is **forced OFF** regardless of the schedule.

BreakTime

After a MaxRuntime stop, the device must remain OFF for this duration before it can turn ON again. This is a mandatory cool-down period.

How They Differ From Cyclic ON/OFF Durations

	Cyclic ON/OFF Duration	MaxRuntime / BreakTime
What is it?	Defines the repeating ON/OFF pattern within a schedule	A global safety limit on continuous operation
Scope	Applies to one specific schedule	Applies to the entire device, across all schedules
Purpose	Operational pattern (e.g., irrigation cycles)	Equipment protection (e.g., motor overheating)
After turning OFF	Next cycle starts after the configured OFF duration	Device must wait for BreakTime before any restart

When MaxRuntime Is Shorter Than Cyclic ON Duration

MaxRuntime will interrupt the cycle early and enforce a BreakTime pause.

Example:

```
MaxRuntime: 20 min | BreakTime: 10 min
Schedule: 09:00–12:00 | Cyclic ON: 30 min, OFF: 30 min

  09:00 ── ON ──► 09:20  MaxRuntime reached → forced OFF
                  09:20 ── BreakTime ──► 09:30 (cool-down complete)
                         ✓ Cycle 2 at 10:00 — CAN START
                  10:00 ── ON ──► 10:20  MaxRuntime reached again → forced OFF
                  ... pattern repeats
```

Each cycle effectively runs for only 20 minutes instead of the configured 30 minutes.

When MaxRuntime Is Longer Than Cyclic ON Duration

MaxRuntime never triggers during normal operation because each cycle ends before the limit is reached.

Example:

```
MaxRuntime: 60 min | BreakTime: 15 min
Schedule: 09:00–12:00 | Cyclic ON: 10 min, OFF: 50 min

  09:00 ── ON ──► 09:10  Cyclic OFF (10 min < 60 min — MaxRuntime not reached)
  10:00 ── ON ──► 10:10  Cyclic OFF
  ... MaxRuntime never triggers
```

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5. Summary Table

Event	Stops Current Cycle?	Remaining Cycles?	Resumes Automatically?
Manual Stop	Yes	All cancelled	No — next schedule occurrence only
Dry Run / Overcurrent	Yes	All blocked	Only if auto-retry is enabled
Low/High Voltage	Yes	Temporarily paused	Yes — clears at next cycle boundary
Phase Failure / Reverse	Yes	Temporarily paused	Yes — clears at next cycle boundary
Float Alert	Yes	Temporarily paused	Yes — clears at next cycle boundary
MaxRuntime Break	Yes	Temporarily paused	Yes — after BreakTime elapses
Cyclic OFF period	Normal operation	Not affected	Yes — next cycle starts automatically
Schedule window ends	Yes	N/A	Next occurrence starts normally

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6. Key Takeaways

1. **Manual stop = full schedule cancel.** If you press stop during a cyclic schedule, all remaining cycles are skipped. The next day's schedule runs normally.
2. **Dry run and overcurrent are treated seriously.** These faults block the entire schedule because they indicate a problem that won't resolve on its own (e.g., a dry pump or a short circuit).
3. **Voltage faults are treated as transient.** They clear between cycles, allowing the device to retry automatically when power conditions normalize.
4. **MaxRuntime is a safety net, not a scheduler.** It protects equipment from running too long. If your cyclic ON duration is shorter than MaxRuntime, they won't interfere with each other.
5. **Auto-retry gives you control over fault recovery.** For each fault type, you can independently enable auto-retry and set the retry delay. This is especially useful for dry run/overcurrent faults in cyclic schedules, where you may want to allow the next cycle to attempt a start.